

05/10/2026

1.1 AI Version Used

ChatGPT, GPT-5.6 Sol

1.2 Date Used

05/10/2026

1.3 How to Help with the Assignment

I asked ChatGPT to describe the assignment and what it would need to complete it. It said the task consisted of the design of a Petri Net, the implementation of a given interface in Java, a set of JUnit tests, and a clear history of development on GitHub.

2.1 AI Version Used

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2.2 Date Used

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2.3 How to Help with the Assignment

I asked ChatGPT to give a simple description of the rail network. It gave simple descriptions of both passenger and freight entries/exits, as well as the reason for the different routes for the two types of trains.

3.1 AI Version Used

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3.2 Date Used

05/10/2026

3.3 How to Help with the Assignment

I asked ChatGPT to define place and transition in the context of Petri Nets. It said that a railway section can be modeled as a place and a movement of a train between two section can be modeled as a transition.

4.1 AI Version Used

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4.2 Date Used

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4.3 How to Help with the Assignment

I asked ChatGPT how to represent section capacity in a Petri Net. It said that a token can be used to represent a train, and that a section can only be occupied by a single train.

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5.1 AI Version Used

5.2 Date Used

06/10/2026

5.3 How to Help with the Assignment

I asked ChatGPT what kind of Java data structure could be used to store what train is in each section. It explained that a `Map<Integer, String>` works well and how the section number can be used as the key.

6.1 AI Version Used

6.2 Date Used

06/10/2026

6.3 How to Help with the Assignment

I asked ChatGPT how another map could be used to store the current position of each train. It explained how a `Map<String, Integer>` can be used to represent a train name to its current track section.

7.1 AI Version Used

7.2 Date Used

06/10/2026

7.3 How to Help with the Assignment

I asked ChatGPT how to check if a train name exists before adding a new train. It provided an example on how `containsKey()` operates with a Java map to determine if a key already exists and why this check is needed before adding a new train.

8.1 AI Version Used

ChatGPT, GPT-5.6 Sol

8.2 Date Used

06/10/2026

8.3 How to Help with the Assignment

I asked ChatGPT how to check if an entry is occupied. It said you have to read the value stored for that section and check if it's null. If it's null, then it's ok to insert the new train.

9.1 AI Version Used

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9.2 Date Used

06/10/2026

9.3 How to Help with the Assignment

I asked ChatGPT what the behavior of the `getSection()` and `getTrain()` methods should be. It said `getSection()` should return the value of the section if the key exists. `getTrain()` should return the value of a train if the key exists. If there's an invalid input, an `IllegalArgumentException` should be thrown.

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10.1 AI Version Used

ChatGPT, GPT-5.6 Sol

10.2 Date Used

07/10/2026

10.3 How to Help with the Assignment

I asked ChatGPT to help with storing a fixed railway route in Java. It said that to represent a route with a series of sections with numbers, like {1, 5, 8}, an integer array can be used.

11.1 AI Version Used

ChatGPT, GPT-5.6 Sol

11.2 Date Used

07/10/2026

11.3 How to Help with the Assignment

I asked ChatGPT how to separate passenger routes from freight routes. It said that for the program to clearly check if a given entry and destination pair is valid, the route map should be separated into two.

12.1 AI Version Used

ChatGPT, GPT-5.6 Sol

12.2 Date Used

07/10/2026

12.3 How to Help with the Assignment

I asked ChatGPT how to find the next section of a train given its route array. It said that to get the next section, the array should be looped through, and the current section should be compared with the following element of the array.

13.1 AI Version Used

ChatGPT, GPT-5.6 Sol

13.2 Date Used

07/10/2026

13.3 How to Help with the Assignment

I asked ChatGPT how to check if a train has reached its final destination. It said that the current section can be compared to the last element of the stored route.

14.1 AI Version Used

ChatGPT, GPT-5.6 Sol

14.2 Date Used

07/10/2026

14.3 How to Help with the Assignment

I asked ChatGPT what should happen after a train reaches its destination. It said that the train should first remain at the destination and depart on the next move request.

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15.1 AI Version Used

ChatGPT, GPT-5.6 Sol

15.2 Date Used

08/10/2026

15.3 How to Help with the Assignment

I asked ChatGPT how should moveTrains() function behave if there are multiple train names in the same call. It said that the function should first determine the intention of each train and then alter the state of the railway.

16.1 AI Version Used

ChatGPT, GPT-5.6 Sol

16.2 Date Used

08/10/2026

16.3 How to Help with the Assignment

I asked ChatGPT to explain why changing the section map immediately inside the movement loop can cause problems. It said later trains can see a partially updated state of that batch and cause inconsistencies.

17.1 AI Version Used

ChatGPT, GPT-5.6 Sol

17.2 Date Used

08/10/2026

17.3 How to Help with the Assignment

I asked ChatGPT how to create a copy of the railway state before processing movements. It said new HashMap<>(sections) can be used to store a copy of the occupancy of the sections.

18.1 AI Version Used

ChatGPT, GPT-5.6 Sol

18.2 Date Used

08/10/2026

18.3 How to Help with the Assignment

I asked ChatGPT how to solve duplicate entries of a train name in the input array. It said LinkedHashSet removes duplicate values and keeps the order of in

19.1 AI Version Used

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19.2 Date Used

08/10/2026

19.3 How to Help with the Assignment

I asked ChatGPT for an explanation of recursion for movement dependency checking. It said a helper method can follow the train occupying the target section and check if that train will also move away.

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20.1 AI Version Used

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20.2 Date Used

09/10/2026

20.3 How to Help with the Assignment

I asked ChatGPT for help in understanding failed autograder logs. It said to compare starting sections, requested movements, end sections, and check for collision, block, or deadlock.

21.1 AI Version Used

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21.2 Date Used

09/10/2026

21.3 How to Help with the Assignment

I asked ChatGPT why a train can be allowed to enter a section that is occupied at the start of the batch. It said that move can be valid if the occupying train is also selected to leave that section in the same batch.

22.1 AI Version Used

ChatGPT, GPT-5.6 Sol

22.2 Date Used

09/10/2026

22.3 How to Help with the Assignment

I asked ChatGPT what should be done if two trains request the same target section. It said they should both wait, as allowing one to move should not be done just because it occurs first in the input.

23.1 AI Version Used

ChatGPT, GPT-5.6 Sol

23.2 Date Used

09/10/2026

23.3 How to Help with the Assignment

I asked ChatGPT how to identify a same-target conflict in Java. It said that a map can be used to count the number of requested movements with the same destination section before the final movements are chosen.

24.1 AI Version Used

ChatGPT, GPT-5.6 Sol

24.2 Date Used

09/10/2026

24.3 How to Help with the Assignment

I asked ChatGPT how to identify cyclic swaps. It said that a set can be used to record trains that are currently checked and returning to a train that is in the set indicates a cycle.

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25.1 AI Version Used

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25.2 Date Used

10/10/2026

25.3 How to Help with the Assignment

I asked ChatGPT how to implement passenger priority at J1. It said that passenger priority should be implemented first to identify an eligible passenger movement and freight movements should be ignored.

26.1 AI Version Used

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26.2 Date Used

10/10/2026

26.3 How to Help with the Assignment

I asked ChatGPT why passenger priority has to be verified before finalizing same-target conflict resolution. It said that when a freight movement is already blocked by priority, it shouldn't be counted as a competitor for the target section.

27.1 AI Version Used

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27.2 Date Used

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27.3 How to Help with the Assignment

I asked ChatGPT how to do basic JUnit 4 tests for this kind of class. It described the use of the annotation `@Test`, the assertions `assertEquals()` and `assertNull()`, and expected exceptions, and I used these examples to write my own tests.

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28.2 Date Used

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28.3 How to Help with the Assignment

I asked ChatGPT what kind of tests would be appropriate for the railway system. It suggested tests for valid and invalid routes, train names, movements, and conflicts; and cases of same-target, simultaneous movement.